

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Correctly identifies the two end points 90 and 270 Condone $\frac{\pi}{2}$ and $\frac{3\pi}{2}$	AO1.2	M1	$90^\circ < x < 270^\circ$
Correctly uses inequalities May state (90, 270) – must be round brackets Again, condone $\frac{\pi}{2}$ and $\frac{3\pi}{2}$	AO2.5	A1	
Total 2 marks			

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Verifies that $x = 1$ gives $y = 3$	AO1.1b	B1	$y = 2 \times 1^2 - 8 \times 1^{\frac{3}{2}} + 8 \times 1 + 1 = 3$ $y = 2x^2 - 8x^{\frac{3}{2}} + 8x + 1$ $\frac{dy}{dx} = 4x - 12\sqrt{x} + 8$ For stationary point $4(\sqrt{x})^2 - 12\sqrt{x} + 8 = 0$ $\sqrt{x} = 1$ or $\sqrt{x} = 2$ $x = 1$ or $x = 4$
	Expresses $x\sqrt{x}$ as $x^{\frac{3}{2}}$	AO1.1b	B1	
	Attempts to differentiate (at least one term correct)	AO1.1a	M1	
	Correctly differentiates	AO1.1b	A1	
	Explains $\frac{dy}{dx} = 0$ for stationary or maximum point Must be explicitly seen	AO2.4	E1	
	Shows solution of $\frac{dy}{dx} = 0$ to give $x = 1$ (and $x = 4$) (May be awarded for work seen in (b)) or correct verification of $x = 1$	AO1.1b	B1	

	Differentiates again (May be awarded for work seen in (b))	AO1.1a	M1	$\frac{d^2 y}{dx^2} = 4 - \frac{6}{\sqrt{x}}$ $x = 1 \text{ gives } \frac{d^2 y}{dx^2} = -2$ Negative so maximum when $x = 1$ Maximum at (1, 3)
	Shows that $x = 1$ gives a negative value (in a correct second differential)	AO1.1b	A1	
	Concludes that maximum point is at (1, 3). Constructs rigorous mathematical argument to show the required result. Failure to score E1 does not rule out award of this mark	AO2.1	R1	
(b)	States coordinates	AO1.1b	B1	(4, 1)
	States minimum point	AO1.1b	B1	Minimum
Total 11 marks				

Notes:

A candidate who does not handle the $x\sqrt{x}$ term can score B1 B0 M1 A0 E1 B0 M1 A0 R0

Q3.

Marking Instructions	AO	Marks	Typical Solution
Identifies and clearly defines consistent variables for length and width. Can be shown on diagram.	AO3.1b	B1	Width of rectangle = $2x$ Length of rectangle = $2y$
Models the area of rectangle with an expression of the correct dimensions	AO3.3	M1	$A = 4xy$
Eliminates either variable to form a model for the area in one variable.	AO1.1a	M1	$x^2 + y^2 = 16$
Obtains a correct equation to model the area in one variable	AO1.1b	A1	$A = 4x\sqrt{16 - x^2}$

Differentiates their expression for area. Condone one error	AO3.4	M1	$\frac{dA}{dx} = 4\sqrt{16-x^2} - \frac{4x^2}{\sqrt{16-x^2}}$
Explains that their derivative equals zero for a maximum or stationary point.	AO2.4	E1	$\frac{dA}{dx} = \frac{64-8x^2}{\sqrt{16-x^2}}$ For maximum point $\frac{dA}{dx} = 0$
Equates area derivative to zero and obtains correct value for either variable. CAO	AO1.1b	A1	$\frac{64-8x^2}{\sqrt{16-x^2}} = 0$ $x = 2\sqrt{2}$
Completes a gradient test or uses second derivative of their area function to determine nature of stationary point	AO1.1a	M1	When $x = 2.8$, $\frac{dA}{dx} = 0.448$ When $x = 2.9$, $\frac{dA}{dx} = -1.191$ Therefore maximum
Deduces that the area is a maximum at $x = 2\sqrt{2}$ or $\theta = \frac{\pi}{4}$ Values need not be exact	AO2.2a	R1	The maximum area is 32 sq in
Obtains maximum area with correct units AWRT 32	AO3.2a	B1	
Total 10 marks			

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Forms expression of the correct form for the gradient of the line AB condone sign error	AO1.1a	M1	Gradient of AB $= \frac{(-4+h)^3 - 48(-4+h) - ((-4)^3 - 48(-4))}{h}$
	Obtains correct expansion of $(-4+h)^3$	AO1.1b	B1	$= \frac{h^3 - 12h^2 + 48h - 64 - 48h + 192 - 128}{h}$
	Obtains correct expansion of numerator	AO1.1b	A1	$= \frac{h^3 - 12h^2}{h}$
	Simplifies numerator and shows given result	AO2.1	R1	$= h^2 - 12h$
(b)	Explains that as $h \rightarrow 0$ the gradient of the line AB $\rightarrow$ the gradient of the curve or tangent to the curve	AO2.4	E1	The gradient of the curve is given by $\lim_{h \rightarrow 0} h^2 - 12h$

Or gradient of curve is given by $\lim_{h \rightarrow 0} h^2 - 12h$ Must not use $h = 0$			
Explains that $\lim_{h \rightarrow 0} h^2 - 12h = 0$ therefore A must be a stationary point	AO2.4	E1	As $h \rightarrow 0$, $h^2 - 12h \rightarrow 0$ therefore A must be a stationary point
Total 6 marks			

Q5.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	7
Total 1 mark			

Q6.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Deduces that the lower bound of x is 1	AO2.2a	M1	$\{x \in \mathbb{R} : x > 1\}$
	States the domain in a correct form	AO2.5	A1	
(b)	Differentiates using quotient rule (condone correct use of product rule) Must have $f'(x) = \frac{(2x-2)^{\frac{1}{2}} - kx(2x-2)^{-\frac{1}{2}}}{(2x-2)}$ OE	AO1.1a	M1	$f'(x) = \frac{(2x-2)^{\frac{1}{2}} - \frac{1}{2}x(2x-2)^{-\frac{1}{2}} \times 2}{(2x-2)}$ $= \frac{2x-2-x}{(2x-2)^{\frac{3}{2}}}$ $= \frac{x-2}{(2x-2)^{\frac{3}{2}}}$
	Obtains correct derivative in unsimplified form	AO1.1b	A1	
	Completes algebraic manipulation, with all previous working correct, to show the correct form. AG	AO2.1	R1	
(c)	States that point of inflection requires second derivative to be 0	AO2.4	E1	For point of inflection $f''(x) = 0$
	Forms an equation $f''(x) = 0$ OE	AO1.1a	M1	
	Solves their equation	AO1.1a	M1	

	Obtains solution $x = 4$	AO1.1b	A1	$f''(x) = \frac{(2x-2)^{\frac{3}{2}} - \frac{3}{2}(x-2)(2x-2)^{\frac{1}{2}} \times 2}{(2x-2)^3}$ $(2x-2)^{\frac{3}{2}} - 3(x-2)(2x-2)^{\frac{1}{2}} = 0$ $(2x-2)^{\frac{1}{2}}[(2x-2) - 3(x-2)] = 0$ $(2x-2)^{\frac{1}{2}}(4-x) = 0$ $x = 1 \text{ or } x = 4$ $x \neq 1 \text{ because of domain}$ $f''(3) = \frac{1}{32} > 0$ $f''(5) = \frac{-\sqrt{2}}{256} < 0$ Therefore point of inflection at $x = 4$
	Gives a valid reason for rejecting $x = 1$, or cancels factor of $(2x - 2)^{1/2}$ stating $x \neq 1$.	AO2.4	E1	
	Tests either side of 'their' $x = 4$	AO1.1a	M1	
	Completes rigorous argument to conclude they have one point of inflection Do not award this mark if 2 nd E1 mark not awarded	AO2.1	R1	
(d)	Deduces values of x for convex section of graph	AO2.2a	B1	$1 < x < 4$
				Total 13 marks

Q7.

Marking Instructions	AO	Marks	Typical Solution								
Obtains $\frac{dy}{dx}$ for both the given curves – at least one term must be correct for each curve	AO3.1a	M1	$\frac{dy}{dx} = 6x^2 + 12x - 12$ $\frac{dy}{dx} = 60 - 12x$ Chris's claim is incorrect when $6x^2 + 12x - 12 \leq 60 - 12x$ $2x^2 + 8x - 24 \leq 0$ $x^2 + 4x - 12 \leq 0$ $(x + 6)(x - 2) \leq 0$ Critical values are $x = -6$ and 2								
States both derivatives correctly	AO1.1b	A1									
Translates problem into an inequality	AO3.1a	M1									
States a correct quadratic inequality FT from an incorrect $\frac{dy}{dx}$ provided both M1 marks have been awarded	AO1.1b	A1									
			<table><tr><td>region</td><td>$x < -6$</td><td>$-6 < x < 2$</td><td>$x \leq 2$</td></tr><tr><td>sign</td><td>+</td><td>-</td><td>+</td></tr></table>	region	$x < -6$	$-6 < x < 2$	$x \leq 2$	sign	+	-	+
region	$x < -6$	$-6 < x < 2$	$x \leq 2$								
sign	+	-	+								

			$-6 \leq x \leq 2$ Chris's claim is incorrect for values of x in the range $-6 \leq x \leq 2$, so he is wrong
Determines a solution to 'their' inequality	AO1.1a	M1	
Obtains correct range of values for x Must be correctly written with both inequality signs correct	AO1.1b	A1	
Interprets final solution in context of the original question, must refer to Chris's claim	AO3.2a	R1	
			Total 7 marks

Q8.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Rewrites given expression with a fractional power and negative power – at least one index form must be correct	AO1.1a	M1	$y = 6x^{\frac{3}{2}} + 32x^{-1}$ $\frac{dy}{dx} = 6 \times \frac{3}{2} \times x^{\frac{1}{2}} - 32x^{-2}$ $= 9\sqrt{x} - \frac{32}{x^2}$
	Both terms correct	AO1.1b	A1	
	Differentiates 'their' rewritten expression – at least one term correct	AO1.1a	M1	
	Both terms correct for 'their' expression	AO1.1b	A1F	
(b)	Finds the equation of the tangent, a clear attempt must be seen	AO3.1a	M1	When $x = 4$, $\frac{dy}{dx} = 9 \times 2 - \frac{32}{16} = 16$ and $y = 6 \times 4 \times 2 + \frac{32}{4} = 56$ Tangent:
	Evaluates 'their' $\frac{dy}{dx}$ (from part (a)) correctly (when $x = 4$)	AO1.1b	A1F	
	Obtains correct y value (when $x = 4$)	AO1.1b	A1	
	Obtains correct form of the equation of a straight line	AO1.1b	A1F	

using 'their' values for y and $\frac{dy}{dx}$			$y - 56 = 16(x - 4)$
Deduces value required at x -axis is when y equals 0 (follow through from 'their' equation) Both coordinates needed, any form	AO2.2a	A1F	When $y = 0$, $x = 4 - \frac{56}{16} = 0.5$ (0.5,0)
Total 9 marks			

Q9.

Marking Instructions	AO	Marks	Typical Solution
Selects an appropriate method – either differentiates, at least as far as: $\frac{dy}{dx} = 2x \dots$ or commences completion of the square: $\left(x - \frac{5}{2}\right)^2 + \dots$	AO1.1a	M1	$y = \left(x - \frac{5}{2}\right)^2 - \frac{25}{4} + a$ y minimised when squared bracket is 0 $\left(\frac{5}{2}, a - \frac{25}{4}\right)$ ALT $\frac{dy}{dx} = 2x - 5$ so $2x - 5 = 0$ for minimum $x = \frac{5}{2}$
Fully differentiates and sets derivative equal to zero or fully completes square Allow one error	AO1.1a	M1	$y = \left(\frac{5}{2}\right)^2 - 5\left(\frac{5}{2}\right) + a = a - \frac{25}{4}$
Obtains both coordinates	AO1.1b	A1	
			Total 3 marks

Q10.

Marking Instructions	AO	Marks	Typical Solution
Explains clearly that $f(x)$ is increasing $\Leftrightarrow f'(x) > 0$ (for all values of x)	AO2.4	E1	For all x , $f'(x) > 0 \Rightarrow f(x)$ is an increasing function $f(x) = x^3 - 3x^2 + 15x - 1$ $\Rightarrow f'(x) = 3x^2 - 6x + 15$

or Explains $\Rightarrow f(x)$ is increasing $f'(x) > 0$ for all values of x This may appear at any appropriate point in their argument			$\Rightarrow f'(x) = 3(x-1)^2 + 12$ $\therefore f'(x)$ has a minimum value of 12 } (*) therefore $f'(x) > 0$ for all values of x
Differentiates – at least two correct terms	AO1.1a	M1	OR for $f'(x)$, $b^2 - 4ac = -144$ $\therefore f'(x) \neq 0$ for any real x , so $f'(x)$ is either always positive or always negative. $f'(0) = 15$
All terms correct	AO1.1b	A1	therefore $f'(x) > 0$ for all values of x
Attempts a correct method which could lead to $f'(x) > 0$	AO3.1a	M1	OR $f''(x) = 6x - 6$, which $= 0$ when $x = 1$ so min $f'(x)$ is $f'(1) = 12$ therefore $f'(x) > 0$ for all values of x
Correctly deduces $f'(x) > 0$ (for all values of x)	AO2.2a	A1	Thus, since, $f'(x) > 0$ for all values of x it is proven that $f(x)$ is an increasing function.
Writes a clear statement that links the steps in the argument together, the deduction about a positive gradient for all values of x proves that the given function is increasing for all values of x	AO2.1	R1	
Total 6 marks			

Q11.

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	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Selects an appropriate routine procedure; evidence of quotient rule or product rule	AO1.1a	M1	$\frac{dy}{dx} = \frac{2(4x^2 + 7) - 8x(2x + 3)}{(4x^2 + 7)^2}$
	Obtains correct derivative (no need for simplification)	AO1.1b	A1	
(ii)	States clearly that $\frac{dy}{dx} > 0 \Rightarrow y$ is increasing	AO2.4	R1	y is increasing $\Leftrightarrow \frac{dy}{dx} > 0$ $\frac{2(4x^2 + 7) - 8x(2x + 3)}{(4x^2 + 7)^2} > 0$
	Forms inequality from	AO3.1a	B1	

'their'			$(4x^2 + 7)^2 > 0$ for all x $\therefore 2(4x^2 + 7) - 8x(2x + 3) > 0$
$\frac{dy}{dx} > 0$			
Deduces numerator must be positive	AO2.2a	R1	$8x^2 + 14 - 16x^2 - 24x > 0$ $4x^2 + 12x - 7 < 0$ (AG)
Considers denominator alone and sets out clear argument to justify given inequality AG Only award this mark if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	
(b) Solves the correct quadratic inequality (accept evidence of factorising, completing the square, use of formula, or correct critical values stated)	AO1.1a	M1	$(2x + 7)(2x - 1)$ $x = -\frac{7}{2}, \frac{1}{2}$
Obtains fully correct answer, given as an inequality or using set notation	AO1.1b	A1	$-\frac{7}{2} < x < \frac{1}{2}$ Or $x \in \left(-\frac{7}{2}, \frac{1}{2}\right)$ Or $x \in \left\{x : -\frac{7}{2} < x < \frac{1}{2}\right\}$
Total 8 marks			

Q12.

Marking Instructions	AO	Marks	Typical Solution
Finds the difference between the maximum and minimum values of y	AO3.1b	M1	$x^2 + 2xy + 2y^2 = 10$ $2x + 2y + 2x \frac{dy}{dx} + 4y \frac{dy}{dx} = 0$
Uses implicit differentiation	AO1.1a	M1	Highest and lowest points occur when $\frac{dy}{dx} = 0$
Differentiates correctly	AO1.1b	A1	
States stationary points occur when $\frac{dy}{dx} = 0$	AO2.4	R1	
			$\frac{dy}{dx} = 0 \Rightarrow x = -y$ $y^2 - 2y^2 + 2y^2 = 10$

Uses $\frac{dy}{dx} = 0$ to find x in terms of y (or vice versa)	AO1.1a	M1	$y = \pm\sqrt{10}$ $\therefore \text{Height} = \sqrt{10} - (-\sqrt{10})$ $= 2\sqrt{10} = 6.32\text{m}$
Finds $x = -y$	AO1.1b	A1	
Deduces maximum and minimum values of y FT 'their' expression provided all M1 marks have been awarded	AO2.2a	A1F	
States the height of the sculpture above the platform FT 'their' max and min values for y provided all M1 marks have been awarded	AO2.2a	A1F	
Total 8 marks			

Q13.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Identifies and clearly defines variables.	AO3.1b	B1	Let C = total cost
	Models the cost with an expression of the form $ax^2 + bxh$	AO3.3	M1	x = length of base edges h = length of height $C = 15x^2 + 32xh$
	Eliminates either variable, using volume equation, to form a model for the cost in one variable.	AO3.3	M1	$x^2h = 60$ $h = \frac{60}{x^2}$
	Obtains a correct equation to model cost in one variable	AO3.1b	A1	$C = 15x^2 + \frac{1920}{x}$ (*)
	Uses their model to find minimum. (at least one term correctly differentiated and equated to zero)	AO3.4	M1	Differentiating $30x - \frac{1920}{x^2} = 0$ $x^3 = 64$ $x = 4$ $h = \frac{60}{4^2} = 3.75$
	Obtains correct equation	AO1.1b	A1	Height of tank is 3.75 m
	Obtains correct value for h	AO3.2a	A1F	$\frac{d^2C}{dx^2} = 30 + \frac{3840}{x^3}$

with correct units in context Award FT from correct substitution into incorrect equation for h but only if all three M1 marks have been awarded, must have correct units.			$x = 4 \Rightarrow \frac{d^2C}{dx^2} > 0$ therefore minimum ALT from (*) onwards $C = 900h^{-1} + 32\sqrt{60}h^{\frac{1}{2}}$ $0 = -900h^{-2} + 16\sqrt{60}h^{-\frac{1}{2}}$
Performs a correct test of 'their' solution: uses the second derivative of 'their' expression for C in terms of x or h to justify that a minimum value for h has been found OE (Second derivative > 0 or test gradient/values either side)	AO2.4	R1	Height of tank is 3.75 m $\frac{d^2C}{dh^2} = 1800h^{-3} - 8\sqrt{60}h^{-\frac{3}{2}}$ $h = 3.75 \Rightarrow \frac{d^2C}{dh^2} = 25.6 > 0$ $\therefore$ minimum

(b)	Explains that two of the sides of the tank will need to have length $x \pm 0.05$ in order to join them	AO3.5c	E1	The sides will need to overlap to be joined, so two of the side lengths will need to be $x + 0.05$
(i)				
(ii)	Explains that the refinement is relatively small and unlikely to have a significant effect on the result.	AO3.5a	R1	The minimum cost is likely to increase slightly, but relative to the size of the tank, an extra 5cm is unlikely to make a significant difference.
Total 10 marks				

Q14.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses a correct method for finding $\frac{dy}{dx}$ evidence for this includes $\frac{dy}{dx}$ or $\frac{dx}{dt}$ and chain rule OR an attempt at implicit or explicit differentiation of a correct Cartesian equation or 'their' equation	AO1.1a	M1	$\frac{dx}{dt} = 3t^2$ $\frac{dy}{dt} = 2t$ $\frac{dy}{dx} = \frac{2t}{3t^2}$ When $t = -2$ $\frac{dy}{dx} = -\frac{1}{3}$ ALT $y = (x-2)^{\frac{2}{3}} - 1$

	from part (b)			
	Obtains correct $\frac{dy}{dx}$	AO1.1b	A1	$\frac{dy}{dx} = \frac{2(x-2)^{-\frac{1}{3}}}{3}$ When $t = -2, x = -6$ $\frac{dy}{dx} = \frac{2(-6-2)^{-\frac{1}{3}}}{3} = -\frac{1}{3}$
	Substitutes $t = -2$ (or $x = -6$) into 'their' equation for $\frac{dy}{dx}$	AO1.1a	M1	
	Obtains correct simplified gradient of the curve FT 'their' equation for $\frac{dy}{dx}$	AO1.1b	A1F	
(b)	Eliminates t or makes t the subject in one expression (evidence for this includes one equation with t as the subject or two equations with equal powers of t .)	AO1.1a	M1	$t^3 = (x - 2), t^2 = (y + 1)$ $t^6 = (x - 2)^2, t^6 = (y + 1)^3$ $(x - 2)^2 = (y + 1)^3$
	Finds a correct Cartesian equation in any form	AO1.1b	A1	ALT $t^3 = (x - 2)$ $t = (x - 2)^{\frac{1}{3}}$ $y = (x - 2)^{\frac{2}{3}} - 1$
Total 6 marks				

Q15.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Uses the product rule for either term	AO1.1a	M1	$y = 2x\cos 3x + (3x^2 - 4)\sin 3x$ $\frac{dy}{dx} = 2\cos 3x - 6x\sin 3x + 6x\sin 3x + 3(3x^2 - 4)\cos 3x$ $= (9x^2 - 10)\cos 3x$
	Uses the product rule for both terms	AO1.1a	M1	
	Differentiates both terms correctly	AO1.1b	A1	
	Rearranges to correct form CAO	AO1.1b	A1	

(b)	Finds $\frac{d^2y}{dx^2}$ from 'their' first derivative and equates to zero	AO3.1a	M1	$\frac{d^2y}{dx^2} = 18x\cos 3x - 3(9x^2 - 10)\sin 3x$
	Applies product rule correctly on $\frac{dy}{dx}$ 'their' $\frac{dy}{dx}$ FT only applies if both M1 marks awarded in part (a)	AO1.1b	A1F	point of inflection $\Rightarrow \frac{d^2y}{dx^2} = 0 \Rightarrow$ $18x\cos 3x - 3(9x^2 - 10)\sin 3x = 0$ $\Rightarrow \frac{\cos 3x}{\sin 3x} = \frac{3(9x^2 - 10)}{18x}$
	Arrives at a result using 'their' second derivative through correct algebraic manipulation that is correct for 'their' second derivative FT only applies if both previous marks in (b) have been awarded.	AO1.1b	A1F	$\Rightarrow \cot 3x = \frac{9x^2 - 10}{6x}$ (AG)
	Constructs a clearly explained rigorous mathematical argument, to show the required result This must include a concluding statement or an explanation of reasoning at the start. AG	AO2.1	R1	
Total 8 marks				

Q16.

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	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds 2 nd derivative and sets up an inequality	AO3.1a	M1	$\frac{dy}{dx} = -2xe^{-x^2}$
	Obtains correct first derivative	AO1.1b	A1	$\frac{d^2y}{dx^2} = -2e^{-x^2} + 4x^2e^{-x^2}$
	Obtains second derivative correct from 'their' first derivative	AO1.1b	A1F	$-2e^{-x^2} + 4x^2e^{-x^2} < 0$ $4x^2 - 2 < 0$
	Deduces correct final inequality (could use set notation)	AO2.2a	A1	$-\frac{\sqrt{2}}{2} < x < \frac{\sqrt{2}}{2}$

(b)	Uses trapezium rule	AO1.1a	M1	$\int_{0.1}^{0.5} e^{-x^2} dx \approx \frac{0.1}{2} (e^{-0.01} + e^{-0.25})$ $+ 2(e^{-0.04} + e^{-0.09} + e^{-0.16})$ ≈ 0.3611
	Trapezium rule entries all correct	AO1.1b	A1	
	Finds correct value	AO1.1b	A1	
(c)	References area being completely within concave section So...	AO2.4	E1	$[0.1, 0.5] \subset \left[-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} \right]$ $\therefore$ area is completely within concave section Hence trapezia lie below curve and give an under-estimate for the area
	Trapezia all fall completely underneath the curve therefore underestimate (only award this mark if previous E1 has been awarded)	AO2.4	E1	
(d)	Uses suitable rectangle to obtain over-estimate	AO3.1a	B1	Using a rectangle with the left hand edge the same height as the curve will produce an over-estimate Area of rectangle = $0.4 \times e^{-0.01^2} = 0.396...$ $\therefore 0.36 < A < 0.40$ So $A = 0.4$ to 1 dp
	Explains that this rectangle lies above the curve	AO2.4	E1	
	Constructs rigorous mathematical argument about accuracy, which leads to required result Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips.	AO2.1	R1	
				Total 12 marks

Q17.

(a) $\left(\frac{dy}{dt} \right) \frac{4t^3}{8} - 2t$

one of these terms correct

M1

all correct (no + c etc)

A1

(b) (i) $t = 1 \Rightarrow \frac{dy}{dt} = \frac{4}{8} - 2$

Correctly sub $t = 1$ into their $\frac{dy}{dt}$

M1

$$= -1\frac{1}{2}$$

must have $\frac{dy}{dt}$ correct (watch for t^3 etc)

A1cso

(ii) $\frac{dy}{dt} < 0$

must have used $\frac{dy}{dt}$ in part (b)(i)

$\Rightarrow$ (height is) **decreasing** (when $t = 1$)

must state that " $\frac{dy}{dt} < 0$ " or " $-1.5 < 0$ " or the equivalent in words

FT their value of $\frac{dy}{dt}$ with appropriate explanation and conclusion

E1✓

(c) (i) $\left(\frac{d^2y}{dt^2} = \right) \frac{4}{8} \times 3t^2 - 2$

Correctly differentiating their $\frac{dy}{dt}$ even if wrongly simplified

M1

$$\left(t = 2, \frac{d^2y}{dt^2} = \right) 4$$

Both derivatives correct and simplified to 4

A1cso

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2

(ii) $\Rightarrow$ minimum

FT their numerical value of $\frac{d^2y}{dt^2}$ from part (c)(i)

E1✓

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[8]

Q18.

(a) (Gradient = $10 - 6 + 5$) = 9

correct gradient from sub $x = 1$ into $\frac{dy}{dx}$

B1

$y - 4 = \text{"their 9"} (x - 1)$
 or $y = \text{"their 9"} x + c$ **and** attempt
 to find c using $x = 1$ and $y = 4$

}

must attempt to use given expression for $\frac{dy}{dx}$ **and** must
 be attempting tangent and not normal **and** coordinates
 must be correct

M1

$$y = 9x - 5$$

condone $y = 9x + c, \dots \quad c = -5$

A1

3

(b) $(y =) \frac{10}{5}x^3 - \frac{6}{3}x^3 + 5x + C$
 one term correct

M1

another term correct

A1

all integration correct including $+ C$

A1

$$4 = 2 - 2 + 5 + C \Rightarrow C = -1$$

substituting both $x = 1$ and $y = 4$ **and** attempting
 to find C

m1

$$y = 2x^5 - 2x^3 + 5x - 1$$

must have $y = \dots$ and coefficients simplified

A1cso

5

[8]

Q19.

(a) $\frac{8}{x^2} = 8x^{-2}$

PI by its derivative as $16x^{-3}$ or $-16x^{-3}$

B1

$$\frac{dy}{dx} = 2 + 16x^{-3}$$

Differentiating either $6 + 2x$ correctly or differentiating $-8/x^2$ correctly.

M1

$$2 + 16x^{-3} \text{ OE}$$

A1

3

(b) At $P(2, 8)$, $\frac{dy}{dx} = 2 + 16 \times 2^{-3} (= 4)$

Attempt to find $\frac{dy}{dx}$ when $x = 2$

M1

Gradient of normal at $P = -\frac{1}{4}$
 $m \times m' = -1$ used

m1

Eqn. of normal at P :

$$y - 8 = -\frac{1}{4}(x - 2) \Rightarrow x + 4y = 34$$

CSO AG

A1

3

(c) (i) At St. Pt $\frac{dy}{dx} = 0$, $2 + 16x^{-3} = 0$

Equating c's $\frac{dy}{dx}$ to 0

Accept ' $\frac{dy}{dx} = 0$ so $x = -2$ ' stated with no errors seen

M1

$$(16x^{-3} = -2) \quad x = -2$$

$$x = -2$$

A1

When $x = -2$, $y = 6 - 4 - 2 = 0$;

$M(-2, 0)$ lies on x -axis

Need statement and correct coords.

A1
3

- (ii) Tangent at M has equation $y = 0$
 $y = 0$ OE

B1
1

- (d) Intersects normal at P when $x + 0 = 34$
PI Solving c's eqn. of tangent with ans (b) as far as correctly eliminating one variable.

M1

$T(34, 0)$
 Accept $x = 34, y = 0$

A1
2

[12]

Q20.

- (a) $f(-3) = (-3)^3 - 4x(-3) + 15$
 $f(-3)$ attempted not long division

M1

$f(-3) = -27 + 12 + 15 = 0 \Rightarrow x + 3$ is a factor
shown = 0 plus statement

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A1
2

- (ii) Quadratic factor $(x^2 - 3x + 5)$
 $-3x$ or $+5$ term by inspection or full long division attempt

$(f(x) =) (x + 3)(x^2 - 3x + 5)$
must see correct product

A1
2

- (b) (i) $\left(\frac{dy}{dx} =\right) 4x^3 - 16x + 60$
one of these terms correct

M1

another term correct

A1

all correct (no +c etc)

A1

3

- (ii) $4x^3 - 16x + 60 = 0$
must see this line OE

$$\Rightarrow x^3 - 4x + 15 = 0$$

AG

B1

1

- (iii) Discriminant of quadratic = $(-3)^2 - 4 \times 5$
discriminant of "their" quadratic or correct use of quad eqn "formula"

M1

$$b^2 - 4ac = -11 \text{ (or } b^2 - 4ac < 0 \text{)}$$

therefore quadratic has no (real) roots

Hence only stationary point is when $x = -3$

correct discriminant evaluated correctly
 (or shown to be < 0) with appropriate conclusion
plus final statement

A1

2

(iv) $\left(\frac{d^2y}{dx^2} = \right) 12x^2 - 16$

B1 ✓

$$= 12(-3)^2 - 16 \quad (\text{or } 12 \times 9 - 16 \text{ etc})$$

sub $x = -3$ into "their" $\frac{d^2y}{dx^2}$

M1

$$= 92$$

A1

3

- (v) Minimum **since** $\frac{d^2y}{dx^2} > 0$ (or $92 > 0$ etc)

*FT appropriate conclusion from their value from
 (iv) plus reason
 treat parts (iv) & (v) holistically*

E1✓

1

[14]

Q21.

(a) $\sqrt{x} = x^{0.5}$

$\sqrt{x} = x^{0.5}$ or $\sqrt{x} = x^{\frac{1}{2}}$ seen or used

B1

$$\frac{12 + x^2\sqrt{x}}{x} = \frac{12 + x^{2.5}}{x}$$

$$= 12x^{-1} + x^{1.5}$$

$12x^{-1}$ or $p = -1$

B1

$x^{1.5}$ or $q = \frac{3}{2}$ (=1.5)

B1

3

(b) (i) $\frac{dy}{dx} = -12x^{-2}$

Ft on c's p only if c's p is a negative integer

B1F

$+ 1.5x^{0.5}$

Ft on c's q only if c's q is a pos non-integer

B1F

2

(ii) When $x = 4$, $y = 11$

B1

When $x = 4$, $\frac{dy}{dx} = \frac{-12}{16} + 3 = \frac{9}{4}$

Attempt to find $\frac{dy}{dx}$ when $x = 4$ PI

M1

Gradient of normal = $-\frac{4}{9}$
 $m \times m' = -1$ used

m1

Eqn of normal: $y - 11 = -\frac{4}{9}(x - 4)$
 ACF eg $4x + 9y = 115$

A1

4

(iii) At St Pt $\frac{dy}{dx} = -12x^{-2} + 1.5x^{0.5} = 0$
 Equating c's $\frac{dy}{dx}$ to zero

M1

$\Rightarrow x^2 x^{0.5} = 8, \Rightarrow x^{\frac{5}{2}} = 8 \Rightarrow x = 8^{\frac{2}{5}}$

A correct eqn in the form $x^n = c$ or $x = c^{\frac{1}{n}}$ correctly obtained.

A1

$\Rightarrow x = (2^3)^{\frac{2}{5}} \Rightarrow x = 2^{\frac{6}{5}}$

CSO $x = 2^{\frac{6}{5}}$. All working must be correct and in an exact form.

If 'x = 0' also appears then A0 CSO

A1

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3

[12]

Q22.

(a) $(y = x^4 \tan 2x)$

$\left(\frac{dy}{dx} =\right) 4x^3 \tan 2x + x^4 2\sec^2 2x$

$4x^3 \tan 2x + Ax^4 \sec^2 kx$

OE where A is a non-zero constant.

M1

A1 for $k = 2$

may have $(\sec 2x)^2$

$$\text{or } \frac{1}{\cos^2 2x}$$

A1

A1 all correct

ISW if attempt to simplify is incorrect.

A1

3

(b) $\left(\frac{dy}{dx}\right) = \frac{\pm 2x(x-1) \pm 1(x^2)}{(x-1)^2}$

Use of the quotient rule

M1

$$\frac{2x(x-1) - 1(x^2)}{(x-1)^2}$$

A1

$$\left(= \frac{x^2 - 2x}{(x-1)^2} \right)$$

Simplification not required

$$\left(\frac{dy}{dx} = \right) \frac{3}{4} \quad \text{or } 0.75 \quad \text{OE}$$

Obtained from correct $\frac{dy}{dx}$

A1

3

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[6]

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Q23.

(a) $\left(\frac{dx}{dt}\right) = -16e^{-2t} \quad \left(\frac{dy}{dt}\right) = 4e^{2t}$

Both derivatives correct

B1

$$\frac{dy}{dx} = \frac{\text{candidate's } \frac{dy}{dt}}{\text{candidate's } \frac{dx}{dt}}$$

chain rule used correctly

M1

$$\frac{dy}{dx} = \frac{4e^{2t}}{-16e^{-2t}} \quad \left(= -\frac{1}{4}e^{4t} \right)$$

Simplification not required $4e^{2t}$ and $-16e^{-2t}$ must be seen.

ISW.

A1

3

- (b) (i) $t = \ln 2$ gradient at $P = -4$
B0 if ISW result is used.

B1ft

1

- (ii) coordinates of P

$$x = -2$$

B1

$$y = 12$$

B1

2

- (iii) gradient of normal = $\frac{1}{4}$
ft gradient at P

B1ft

$$\text{equation of normal } \frac{y-12}{x-(-2)} = \frac{1}{4}$$

Set up equation of normal

M1

$$\text{at } y = 0 \quad x = -50$$

$$(-50, 0) \text{ CSO}$$

A1

3

- (c) $xy + 4y - 4x = (8e^{-2t} - 4)(2e^{2t} + 4) + 4(2e^{2t} + 4) - 4(8e^{-2t} - 4)$
Write $xy + 4y - 4x$ in terms of t .

M1

$$= 16 + 32e^{-2t} - 8e^{2t} - 16 + 8e^{2t} + 16 - 32e^{-2t} + 16$$

Multiply out and simplify using $e^{-2t}e^{2t} = 1$ PI

m1

$$(xy + 4y - 4x) = 32$$

Correct working to $k = 32$

$k = 32$ NMS; SC1

A1

3

Alternative:

$$e^{-2t} = \frac{x+4}{8} \text{ or } e^{2t} = \frac{y-4}{2}$$

Write e^{-2t} in terms of x or e^{2t} in terms of y .
Condone sign errors

(M1)

$$e^{-2t}e^{2t} = \left(\frac{x+4}{8}\right)\left(\frac{y-4}{2}\right) = \frac{xy+4y-4x-16}{16} = 1$$

Multiply out and use $e^{-2t} e^{2t} = 1$

(m1)

$$xy + 4y - x = 32$$

All correct with $k = 32$

(A1)

(3)

[12]

Q24.

(a) (i) $\left(\frac{dy}{dx} = \right) 5x^4 - 6x + 1$
one term correct

M1

another term correct

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A1

all correct (no + c etc)

A1

(ii) $\left(\frac{d^2y}{dx^2} = \right) 20x^3 - 6$
FT 'their' $\frac{dy}{dx}$

B1✓

(b) $x = -1 \Rightarrow \frac{dy}{dx} = 5(-)^4 - 6(-1) + 1 (= 12)$
must sub $x = -1$ into 'their' $\frac{dy}{dx}$

M1

$$\Rightarrow y = 12(x + 1)$$

any correct form with $(x - -1)$ simplified
condone $y = 12x + c$, $c=12$

A1cso

(c) $x = 1 \Rightarrow \frac{dy}{dx} = 5 - 6 + 1$

sub $x = 1$ into their $\frac{dy}{dx}$

$$\frac{dy}{dx} = 0 \Rightarrow \text{stationary point}$$

shown = 0 plus correct statement

A1cso

when $x = 1$, $\frac{d^2y}{dx^2} = 14 \Rightarrow (B \text{ is a }) \text{ minimum (point)}$

or $\frac{d^2y}{dx^2} = 20 - 6 > 0 \Rightarrow (B \text{ is a }) \text{ minimum (point)}$

must have correct $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for E1

E1

(d) (i) $\frac{x^6}{6} - \frac{3x^3}{3} + \frac{x^2}{2} + 5x$

one term correct

M1

another term correct

A1

all correct (may have + c)

A1

$$\left[\frac{1}{6} - 1 + \frac{1}{2} + 5 \right] - \left[\frac{1}{6} + 1 + \frac{1}{2} - 5 \right]$$

'their' $F(1) - F(-1)$ with powers of 1 and
-1 evaluated correctly

M1

$$= 8$$

A1cso

(ii) 'their answer to part (i)' – 2

M1

$$\Rightarrow \text{Area} = 6$$

A1cso

2

[16]

Q25.

(a) $\frac{dy}{dx} = 12 - 5x^{\frac{2}{3}}$

$kx^{\frac{2}{3}}$ term.

M1

ACF

A1

2

(b) (i) When $x = 0$, $\frac{dy}{dx} = 12$

Ft on c's y' evaluated correctly at $x = 0$

B1F

Eqn of tangent at O is $y = 12x$

OE Ft on c's value for $y'(0)$ provided $y'(0) > 0$.

B1F

2

(ii) When $x = 8$, $\frac{dy}{dx} = 12 - 5 \times (8)^{\frac{2}{3}}$

Attempt to find $\frac{dy}{dx}$ when $x = 8$

M1

Equation of tangent at $(8, 0)$ is $y - 0 = y'(8)[x - 8]$

$y = y'(8)[x - 8]$ OE

M1

$$y = -8(x - 8) \Rightarrow y + 8x = 64$$

CSO AG

A1

3

(c) $\int \left(12x - 3x^{\frac{5}{3}} \right) dx = \frac{12x^2}{2} - \frac{3x^{\frac{8}{3}}}{\frac{8}{3}} (+c)$
 $kx^{\frac{5}{3}+1}$ term after integrating, condone k left unsimplified for this M mark.

$= 6x^2 - \frac{9}{8}x^{\frac{8}{3}} (+c)$
For $6x^2$ OE eg $(12x^2 / 2)$

B1

For $-\frac{9}{8}x^{\frac{8}{3}}$ OE

A1

3

(d) Area bounded by curve and x -axis

$= \int_0^8 \left(12x - 3x^{\frac{5}{3}} \right) dx = 6 \times 8^2 - \frac{9}{8} \times (8)^{\frac{8}{3}}$
 $\pm F(8) \{- F(0)\}$ PI following integration

M1

$= 384 - 288 = 96$

PI by correct final answer if evaluation not seen here

A1

At P , $12x + 8x = 64$

Solving $y + 8x = 64$ and c 's $y = kx$, $k > 0$, down to an eqn in one variable...

$[y + 2y / 3 = 64]$

M1

$(x_p = 3.2) \quad y_p = 38.4$

For $y_p 38.4$ OE [If using integration to find area of triangle, award A1 if both ' $x_p = 3.2$ ' and correct integration of correct eqns of the 2 lines]

A1

Area of triangle $OPA = \frac{1}{2} \times 8 \times y_p$

OE Need perpendicular ht to be linked to $y_p > 0$.

M1

Area of shaded region = Area ΔOPA - $\int_0^8 \left(12x - 3x^{\frac{5}{3}} \right) dx$
 M0 if evaluated to a value < 0

M1

$= 153.6 - 96 = 57.6$

OE eg 288 / 5

A1

7

[17]

Q26.

(a) $xy^2 + 3y = (8t^2 - t) \left(\frac{3}{t} \right)^2 + 3 \left(\frac{3}{t} \right)$
 Substitute and expand

M1

$= 72 - \frac{9}{t} + \frac{9}{t} = 72$
 $k = 72$

A1

2

Alternative

$x = 8 \left(\frac{3}{y} \right)^2 - \frac{3}{y}$

Eliminate t

(M1)

$xy^2 + 3y = 72$

$k = 72$

(A1)

(2)

(b) (i) $\frac{dx}{dt} = 16t - 1 \quad \frac{dy}{dt} = -\frac{3}{t^2}$

B1B1

$$t = \frac{1}{4} \quad \frac{dy}{dx} = \frac{-\frac{3}{(\frac{1}{4})^2}}{16 \times \frac{1}{4} - 1}$$

Use chain rule $\left(\frac{dy}{dx} = \frac{-3}{16t^3 - t^2} \right)$ and calculate gradient using $t = \frac{1}{4}$

M1

$$= -16$$

A1

$$t = \frac{1}{4} \quad x = \frac{8}{16} - \frac{1}{4} \quad y = \frac{3}{\frac{1}{4}}$$

Calculate x and y using $t = \frac{1}{4}$

M1

$$x = \frac{1}{4} \quad y = 12$$

Both correct

A1

tangent $y = -16x + 16$
ACF CSO

$$y - 12 = -16 \left(x - \frac{1}{4} \right) \text{ ISW}$$

A1

7

Alternative

$$2xy \frac{dy}{dx} + y^2$$

Product rule attempted; two terms added, one with $\frac{dy}{dx}$

(M1A1)

$$+ 3 \frac{dy}{dx} = 0$$

(B1)

$$t = \frac{1}{4} \quad x = \frac{8}{16} - \frac{1}{4} \quad y = \frac{3}{\frac{1}{4}}$$

Calculate x and y using $t = \frac{1}{4}$

(M1)

$$x = \frac{1}{4} \quad y = 12$$

Both correct.

(A1)

$$\left(\frac{dy}{dx} = \frac{-y^2}{2xy + 3} \right) \frac{dy}{dx} = -16$$

Calculate gradient from candidate's expression.

m1

$$\text{tangent } y = -16x + 16$$

ACF CSO

$$y - 12 = -16 \left(x - \frac{1}{4} \right) \quad \text{ISW}$$

(A1)

(7)

Alternative

$$x = \frac{72 - 3y}{y^2}$$

Correct expression for x from candidate's implicit equation.

Quotient rule attempted; y^4 and two terms subtracted.

(M1)

$$\frac{dx}{dy} = \frac{y^2(-3) - (72 - 3y) \times 2y}{y^4}$$

(A1)

$$\left(\frac{dx}{dy} = \frac{3y - 144}{y^3} \right)$$

Numerator; first term; second term

(A1)

$$t = \frac{1}{4} \quad y = \frac{3}{\frac{1}{4}} = 12$$

(B1)

$$\frac{dx}{dy} = -\frac{1}{16} \quad \frac{dy}{dx} = -16$$

Use $t = \frac{1}{4}$ to calculate y

m1

$$x = \frac{1}{4} \quad x = \frac{8}{16} - \frac{1}{4} = \frac{1}{4}$$

Evaluate and invert.

(B1)

$$y = -16x + 16$$

(A1)

(7)

Alternative for $\frac{dx}{dy}$
Use $t = \frac{1}{4}$ to calculate x

$$x = \frac{72}{y^2} - \frac{3}{y}$$

ACF CSO

(M1)

$$\frac{dx}{dy} = -\frac{144}{y^3} + \frac{3}{y^2}$$

(A1)

$$\left(\frac{dx}{dy} = \frac{3y - 144}{y^3} \right)$$

Correct expression for x from candidate's implicit equation and attempt derivatives

(A1)

$$(ii) \quad y = -16 \times \frac{3}{2} + 16 = -8$$

Substitute $x = \frac{3}{2}$ into candidate's tangent; calculate y

M1

$$\frac{3}{2}(-8)^2 + 3 \times (-8) = 96 - 24 = 72$$

$y = -8$ used to verify 72

A1

2

[11]

Q27.

- (a) (i) $3x^2 + 3x^2 + xy + xy + 3xy + 3xy$
correct expression for surface area

M1

$$6x^2 + 8xy = 32 \Rightarrow 3x^2 + 4xy = 16$$

$$2(3x^2 + xy + 3xy) = 32 \text{ etc}$$

AG be convinced

A1

2

- (ii) $(V =) 3x^2 y$ OE
correct volume in terms of x and y

M1

$$= 3x \left(\frac{16 - 3x^2}{4} \right) \text{ or } = 3x^2 \left(\frac{16 - 3x^2}{4x} \right)$$

OE

$$= 12x - \frac{9x^3}{4}$$

CSO AG

be convinced that all working is correct

A1

2

(b) $\left(\frac{dV}{dx} = \right) 12 - \frac{27}{4} x^2$

one of these terms correct

M1

all correct with 9×3 evaluated (no + c etc)

A1

2

(c) (i) $x = \frac{4}{3} \Rightarrow \frac{dV}{dx} = 12 - \frac{27}{4} \times \left(\frac{4}{3}\right)^2$
attempt to sub $x = \frac{4}{3}$ into 'their' $\frac{dV}{dx}$

M1

$$\frac{dV}{dx} = 12 - \frac{27}{4} \times \frac{16}{9} = 12 - 12$$

or $12 - \frac{432}{36} = 12 - 12$ or $12 - \frac{48}{4} = 0$ etc

$$\frac{dV}{dx} = 0 \Rightarrow \text{stationary value}$$

CSO; shown = 0 plus statement

A1

2

(ii) $\frac{d^2V}{dx^2} = -\frac{27x}{2}$ OE
FT for 'their' $\frac{dV}{dx} = a + bx^2$

B1 ✓

when $x = \frac{4}{3}$, $\frac{d^2V}{dx^2} < 0 \Rightarrow \text{maximum}$
or sub of $x = \frac{4}{3}$ into 'their' $\frac{d^2V}{dx^2} \Rightarrow \text{maximum}$

$\left(\text{FT "minimum" if their } \frac{d^2V}{dx^2} > 0 \right)$
 EO if numerical error seen

E1 ✓

2

[10]

Q28.

(a) (i) $\left. \begin{array}{l} (\text{Increasing} \Rightarrow) \frac{dy}{dx} > 0 \\ 20x - 6x^2 - 16 > 0 \end{array} \right\} \text{ either}$
correct interpretation of y increasing

M1

$$\Rightarrow 6x^2 - 20x + 16 < 0 \quad \text{or} \quad (2) \quad (10x - 3x^2 - 8) > 0$$

must see at least one of these steps before final answer for A1

$$\Rightarrow 3x^2 - 10x + 8 < 0$$

CSO AG no errors in working

A1

2

(ii) $(3x - 4)(x - 2)$

correct factors or correct use of quadratic equation

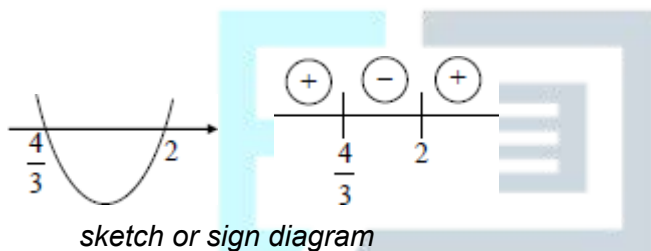
formula as far as $\frac{10 \pm \sqrt{4}}{6}$

M1

CVs are $\frac{4}{3}$ and 2

condone $\frac{8}{6}$ and $\frac{12}{6}$ here but not in final line

A1



M1

$$\frac{4}{3} < x < 2$$

$$\text{or } 2 > x > \frac{4}{3}$$

$$\text{accept } x < 2 \text{ AND } x > \frac{4}{3}$$

$$\text{but not } x < 2 \text{ OR } x > \frac{4}{3}$$

$$\text{nor } x < 2, x > \frac{4}{3}$$

A1

Mark their final line as their answer

4

(b) (i) $x = 2; \left(\frac{dy}{dx} = \right) 40 - 24 - 16$

sub $x = 2$ into $\frac{dy}{dx}$ and simplify terms

M1

$\frac{dy}{dx} = 0 \Rightarrow$ tangent at P is parallel to the x -axis
 must be all correct working plus statement

A1

2

(ii) $x = 3$; $\frac{dy}{dx} = 20 \times 3 - 6 \times 3^2 - 16$
 must attempt to sub $x = 3$ into $\frac{dy}{dx}$

M1

$$(= 60 - 54 - 16) = -10$$

A1

Gradient of normal = $\frac{1}{10}$

$$\frac{-1}{\text{"their"} -10}$$

A1 ✓

Normal: $(y - -1) = \text{'theirgrad'}(x - 3)$
 normal attempted with correct coordinates

used and gradient obtained from their $\frac{dy}{dx}$ value

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$y + 1 = \frac{1}{10}(x - 3)$
 any correct form, eg $10y = x - 13$ but
 must simplify -- to +

A1

(Equation of tangent at P is) $y = 3$

B1

$$x = 43$$

CSO; $\Rightarrow R(43, 3)$

A1

Q29.

- (a) (i) (When $x = 2$) $\frac{dy}{dx} = 12 - 1 - 11 = 0$
AG Must see intermediate evaluations

B1

1

- (ii) $\frac{4}{x^2} = 4x^{-2}$ {so $\frac{dy}{dx} = 3x^2 - 4x^{-2} - 11$ }
 $\frac{4}{x^2} = 4x^{-2}$, seen in (a)(ii) or earlier. PI by $\pm 8x^{-3}$ term in answer

B1

$$\frac{d^2y}{dx^2} = 6x + 8x^{-3}$$

Correct powers of x correctly obtained from differentiating the first two terms

M1

$$6x + 8x^{-3} \text{ ACF}$$

A1

When $x = 2$, $\frac{d^2y}{dx^2} = 12 + 8/8 = 13$

A1

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4

- (iii) Since $\frac{d^2y}{dx^2} > 0$, P is a minimum point.
Ft on c's value of $y''(2)$ in (a)(ii) but must see reference to sign of $y''(2)$ either explicitly or as inequality, as well as the correct ft conclusion

E1F

1

(b) $\int \left(3x^2 - \frac{4}{x^2} - 11 \right) dx = x^3 + 4x^{-1} - 11x (+c)$

Attempt to integrate $\frac{dy}{dx}$ with at least two of the three terms integrated correctly

M1

$$(y =) x^3 + 4x^{-1} - 11x(+c)$$

For $x^3 + 4x^{-1} - 11x$ OE even unsimplified.

1

When $x = 2, y = 1 \Rightarrow 1 = 8 + 2 - 22 + c$

Substituting

$x = 2, y = 1$ into $y = F(x) + 'c'$ in attempt to find constant of integration, where $F(x)$ follows attempted integration of $\frac{dy}{dx}$ expression for $\frac{dy}{dx}$

$$y = x^3 + 4x^{-1} - 11x + 13$$

ACF

A1

4

[10]

Q30.

(a) $\left(\frac{dy}{dx} =\right) x^3 \times \frac{1}{x} + 3x^2 \ln x$

$$px^3 \times \frac{1}{x} + qx^2 \ln x$$

where p and q are integers

M1

$$p = 1, q = 3$$

A1

2

(b) (i) $\left(\frac{dy}{dx} =\right) e^2 + 3e^2 \ln e \quad (= 4e^2)$

Substituting e for x in their $\frac{dy}{dx}$, but must have scored M1 in (a)

M1

$$y = e^3 \ln e (= e^3)$$

B1

$$y - e^3 = 4e^2 (x - e)$$

OE **but** must have evaluated $\ln e$ (twice) for this mark (must be in exact form, but condone numerical evaluation after correct equation)

A1

3

(ii) $-e^3 = 4e^2(x - e)$ or $4e^2x = 3e^3$ OE

Correctly substituting $y = 0$ into a correct tangent equation in (b)(i)

M1

$$x = \frac{3}{4}e$$

CSO;

ignore subsequent decimal evaluation

A1

2

[7]

Q31.

(a) $\left(\frac{dx}{dy}\right) = \frac{\cos y \times \cos y - \sin y \times -\sin y}{\cos^2 y}$

Condone incorrect signs, poor notation, omission of $\frac{dy}{dx}$ or using $\frac{dx}{dy}$

M1

$$= \frac{\cos^2 y + \sin^2 y}{\cos^2 y}$$

*RHS correct with terms squared, including correct notation
Must see this line*

A1

$$= \frac{1}{\cos^2 y} \quad \text{or } (= 1 + \tan^2 y)$$

Must see one of these

$$\frac{dx}{dy} = \sec^2 y$$

AG; all correct including correct use of $\frac{dx}{dy}$ throughout

A1
CSO

3

(b) $\sec^2 y = 1 + (x - 1)^2$

Correct use of $\sec^2 y = 1 + \tan^2 y$ and in terms of x

M1

$$= 1 + x^2 - 2x + 1 \quad \text{OE}$$

$$= x^2 - 2x + 2$$

AG; must see “ $\sec^2 y =$ ”, $(x - 1)^2$ expanded and no errors seen

A1

2

(c) $\frac{dx}{dy} = x^2 + 2x + 2$ or $\frac{dy}{dx} = \frac{1}{\sec^2 y}$
Must be seen

$$\frac{dy}{dx} = \frac{1}{x^2 - 2x + 2}$$

AG and no errors seen

B1

1

(d) (i) $y = \tan^{-1}(x - 1) - \ln x$

$$\left(\frac{dy}{dx} \right) = \frac{1}{x^2 - 2x + 2} - \frac{1}{x}$$

Must be correct

M1

$$\left(\frac{dy}{dx} = 0 \right)$$

$$\pm x^2 + bx + c (= 0)$$

Expression in this form (generous), where b and $c \neq 0$

m1

$$x^2 - 3x + 2 = 0$$

Must see correct equation = 0

A1

$$x = 1, 2$$

Both answers must be seen

The two A marks are independent

A1

4

(ii) $y'' = p(x^2 - 2x + 2)^{-2}(2x - 2) \pm qx^{-2}$

where p and q are constants

M1

$$y'' = -(x^2 - 2x + 2)^{-2} (2x - 2) + x^{-2}$$

$p = -1, q = 1$ including correct brackets

A1

2

(iii) $x = 1, y'' = 1$

Must have scored full marks in (d)(i) and (ii)

M1

At $x = 1, y'' > 0 \therefore \min$

When $x = 1, y = 0$ hence on x -axis

Must see $y'' > 0$ or in words

Both statements fully correct

A1

2

[14]

Q32.

(a) (i) $\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{6\cos 2\theta}{-2\sin \theta}$

condone coefficient errors

M1

A1

$$= \frac{6(1 - 2\sin^2 \theta)}{-2\sin \theta}$$

Use $\cos 2\theta = 1 - 2\sin^2 \theta$

m1

$$= 6\sin \theta - 3\operatorname{cosec} \theta$$

$$a = 6 \quad b = -3$$

A1

4

(ii) $\theta = \frac{\pi}{6} \quad \frac{dy}{dx} = 6 \times \frac{1}{2} - 3 \times 2 = -3$

$\theta = \frac{\pi}{6}$ substituted into their $\frac{dy}{dx}$ and evaluated

B1ft

gradient normal = $\frac{1}{3}$
 ft $\frac{dy}{dx}$, provided non-zero

B1ft

2

(b)

$$\left. \begin{aligned} y &= 6 \sin \theta \cos \theta \\ &= (\pm) 6 \sqrt{1 - \cos^2 \theta} \times \cos \theta \\ &= (\pm) 6 \sqrt{1 - \left(\frac{x}{2}\right)^2} \times \left(\frac{x}{2}\right) \end{aligned} \right\}$$

Correct expansion of $\sin 2\theta$ and use $x = 2\cos\theta$ to eliminate θ

M1

Correct elimination of θ

A1

$$y^2 = \frac{9}{4}x^2(4 - x^2)$$

$$p = \frac{9}{4} \text{ OE and } (4 - x^2) \text{ shown}$$

A1

Alternative: using verification

$$y^2 = 9\sin^2 2\theta = 36\sin^2 \theta \cos^2 \theta$$

must be squared

(M1)

$$x^2(4 - x^2) = 4\cos^2 \theta \times 4\sin^2 \theta$$

(A1)

$$p = \frac{9}{4} \quad \text{OE}$$

$$\text{or } y^2 = \frac{9}{4}x^2(4 - x^2)$$

(A1)

3

[9]

Q33.

$$9x^2 - 6xy + 4y^2 = 3$$

$$18x = 0$$

$$= 0 \text{ PI}$$

B1

$$-6y - 6x \frac{dy}{dx}$$

$$\text{or } \frac{d(6xy)}{dx} = 6y + 6x \frac{dy}{dx} \text{ seen separately}$$

B1

$$+ 8y \frac{dy}{dx}$$

$$\frac{dy}{dx}(-6x + 8y) = 6y - 18x$$

B1

$$\text{Use } \frac{dy}{dx} = 0$$

M1

$$\Rightarrow y = 3x \text{ or } x = \frac{y}{3}$$

CSO

A1

$$y = 3x \Rightarrow 9x^2 - 6x \times 3x + 4(3x)^2 = 3$$

$$27x^2 = 3 \Rightarrow x = \pm \frac{1}{3} \text{ OE}$$

Substitute $y = ax$ into equation and solve for a value of x or y . Condone missing brackets.

m1

*Both values of x or y required.
ft on their $y = 3x$*

A1ft

$$\left(\frac{1}{3}, 1\right) \quad \left(-\frac{1}{3}, -1\right)$$

*CSO Correct corresponding simplified values of x and y .
Withhold if additional answers given*

A1

Q34.

(a) $\frac{dy}{dx} = 18 + 6x - 12x^2$

one of these terms correct

M1

another term correct

A1

all correct (no + c etc)

(penalise + c once only in question)

A1

3

(b) $18 + 6x - 12x^2 = 0$

putting their $\frac{dy}{dx} = 0$,

PI by attempt to solve or factorise

M1

$6(3 - 2x)(x + 1) (= 0)$

*attempt at factors of **their** quadratic
or use of quadratic equation formula*

m1

$x = -1, x = \frac{3}{2}$ OE

*must see both values unless $x = -1$
is verified separately*

A1

3

If M1 not scored, award SC B1 for verifying

that $x = -1$ leads to $\frac{dy}{dx} = 0$ and a further

SC B2 for finding $x = \frac{3}{2}$ as other value

(c) (i) $\frac{d^2y}{dx^2} = 6 - 24x$

FT their $\frac{dy}{dx}$ but $\frac{d^2y}{dx^2}$ must be correct if

3 marks earned in part (a)

B1 ✓

When $x = -1$, $\frac{d^2y}{dx^2} = 6 - (24 \times -1)$

Sub $x = -1$ into their $\frac{d^2y}{dx^2}$

M1

$$\frac{d^2y}{dx^2} = 30$$

A1cso

3

(ii) Minimum point

must have a value in (c)(i)

FT "maximum" if their value of $\frac{d^2y}{dx^2} < 0$

E1 ✓

1

[10]

Q35.

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(a) (i) $\frac{dy}{dx} = -1 - 4x^3$

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one of these terms correct

M1

all correct (no + c)

A1

(When $x = 1$, grad =) -5

(Check that $\frac{dy}{dx}$ is actually correct!)

A1cso

3

(ii) $y - 12 = \text{'their grad'} (x - 1)$

any form of equation through (1, 12) and
attempt at c if using $y = mx + c$

M1

$$y = -5x + 17 \text{ (or } y = 17 - 5x)$$

FT their gradient

Condone $y = -5x + c$, $c = 17$ etc

A1 ✓

2

(b) (i) $14x - \frac{x^2}{2} - \frac{x^5}{5}$

one of these terms correct

M1

another term correct

A1

all correct (may have $+c$)

A1

$$[]_2 =$$

$$\left(14 - \frac{1}{2} - \frac{1}{5}\right) - \left(-28 - 2 + \frac{32}{5}\right)$$

$F(1)$ and $F(-2)$ attempted

m1

$$= 36.9 \quad \text{OE}$$

Condone recovery to this value

A1

5

(ii) Area $\Delta = \frac{1}{2} \times 3 \times 12$

Correct area of triangle unsimplified

M1

$$= 18$$

$$\Rightarrow \text{shaded area} = 18.9$$

A1cso

2



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